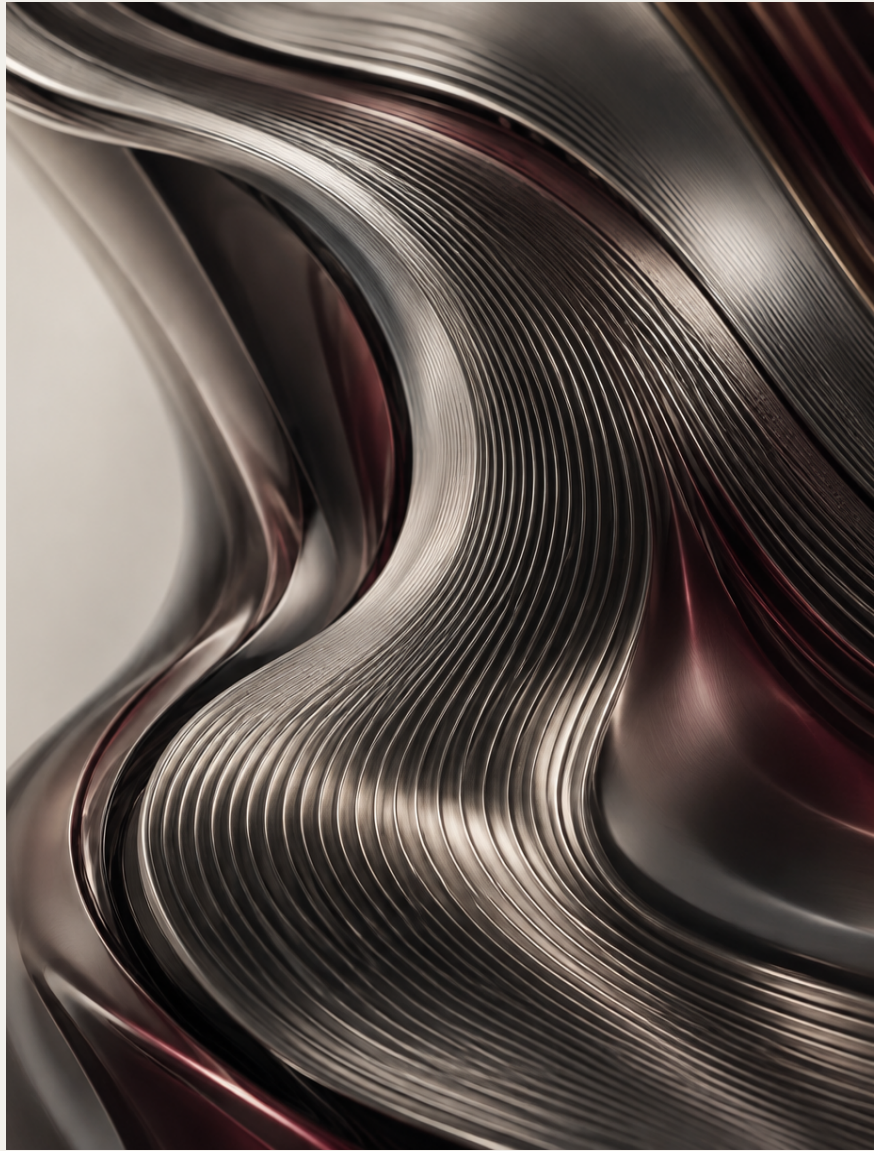


AI ENGINEER & FOUNDER

Make it
mean
something.

Software that turns information
into useful tools.



Nikhil
Yarra.

Building Teyrin.

Intelligence, put to work.

[View the live catalogue](#)

From data to applied AI.



I started in telecom, healthcare and financial data, then moved into retrieval, conversational agents and multimodal AI.

My work spans APIs, tool use, validation, fallbacks and evaluation. I focus on building useful software around the model.

M.S. Data Science

New Jersey Institute of Technology

B.Tech. Computer Science & Engineering

GITAM, Hyderabad

WHAT I CAN HELP BUILD

Document intelligence

Retrieval, source inspection and evaluation.

Workflow software

APIs, automation and human review paths.

Web applications

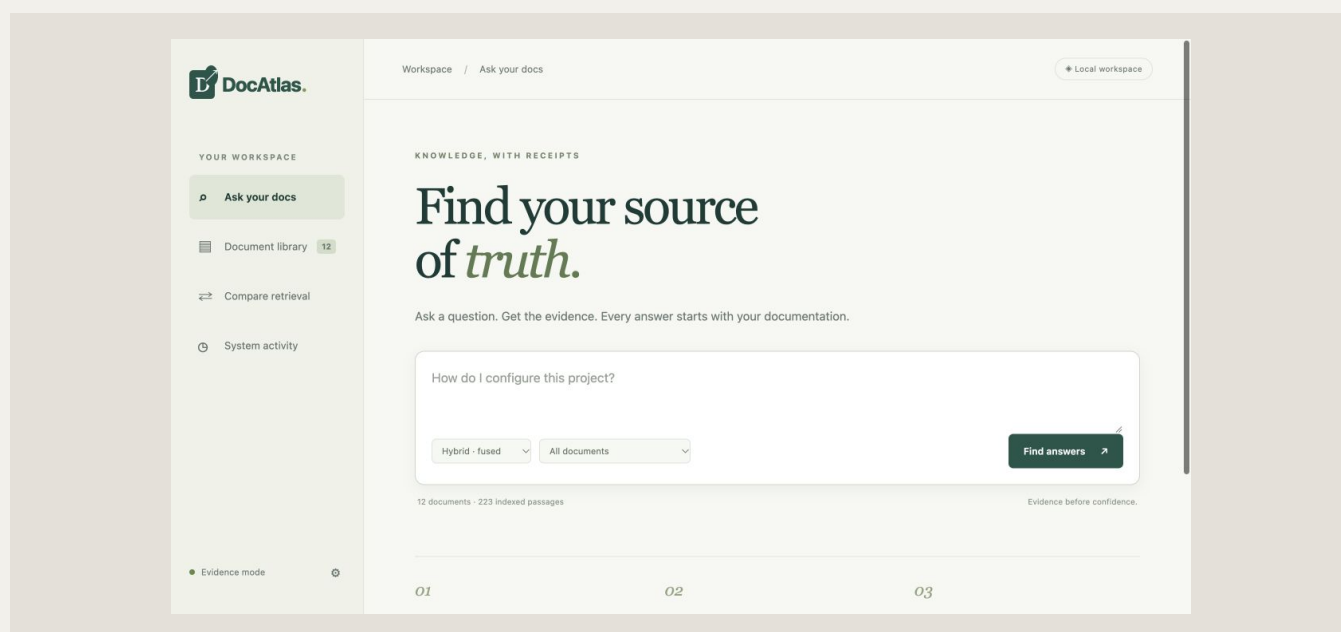
Practical interfaces for complex information.

DocAtlas

Documentation answers with inspectable evidence.

DEVELOPMENT SYSTEM

FastAPI / SQLite / BM25 / embeddings



Original repository workspace capture.

PURPOSE

Find answers across technical documents while keeping every response connected to its sources.

BUILD

Markdown, text, HTML and text-based PDFs feed section-aware ingestion, SQLite FTS5 and optional local embeddings. Search supports BM25, neural retrieval and hybrid rank fusion.

EVIDENCE & SCOPE

A development evaluation used 12 public FastAPI pages, 223 passages and 30 AI-authored questions. BM25 and hybrid reached 95.83% evidence recall@5; BM25 had the highest MRR@5 at 0.809. These are retrieval results, not answer accuracy. Live generation providers are not yet validated. Quote checks verify source membership and exact quotation, not semantic entailment. The service is a single workspace, not a multi-tenant SaaS.

Inspect repository

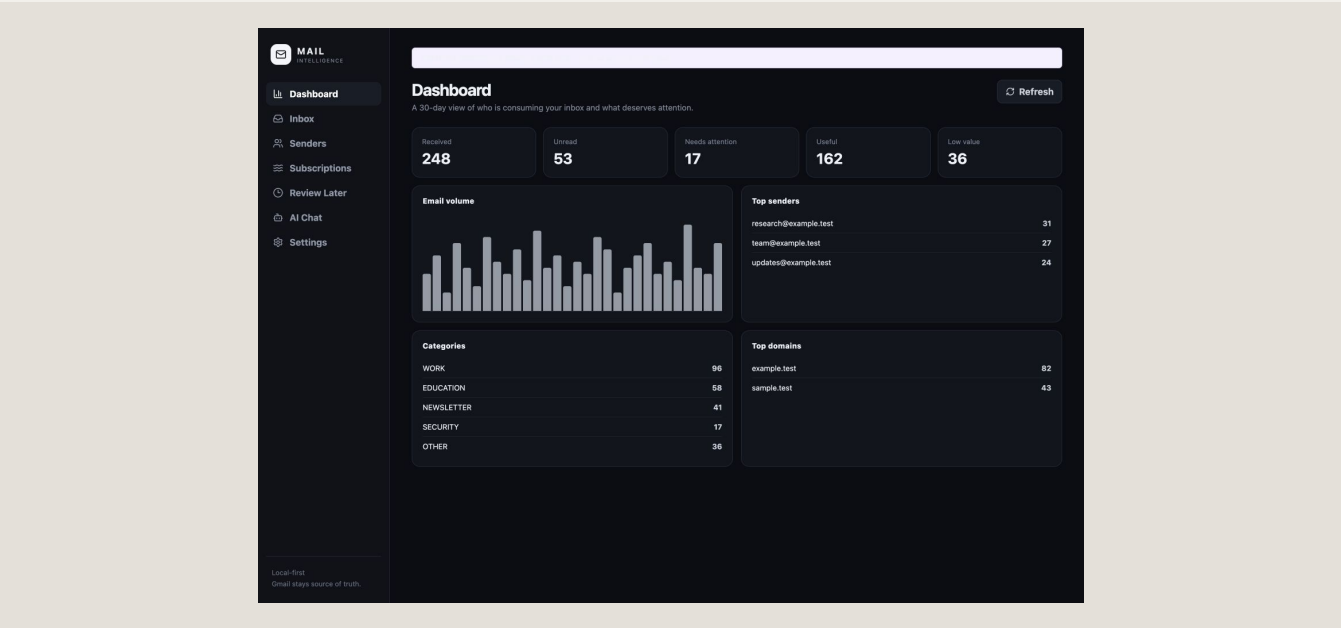
Reviewed source revision

Mailayer

Search an inbox. Keep the source in view.

LOCAL PROTOTYPE

React / FastAPI / SQLite FTS5 / local models



Original frontend with labelled synthetic data.

- PURPOSE

Connect mailbox search, priorities and answers in one interface.
- BUILD

A read-only Gmail gateway prepares messages for local SQLite storage. Keyword search can combine with semantic retrieval; chat packages message evidence alongside responses.
- EVIDENCE & SCOPE

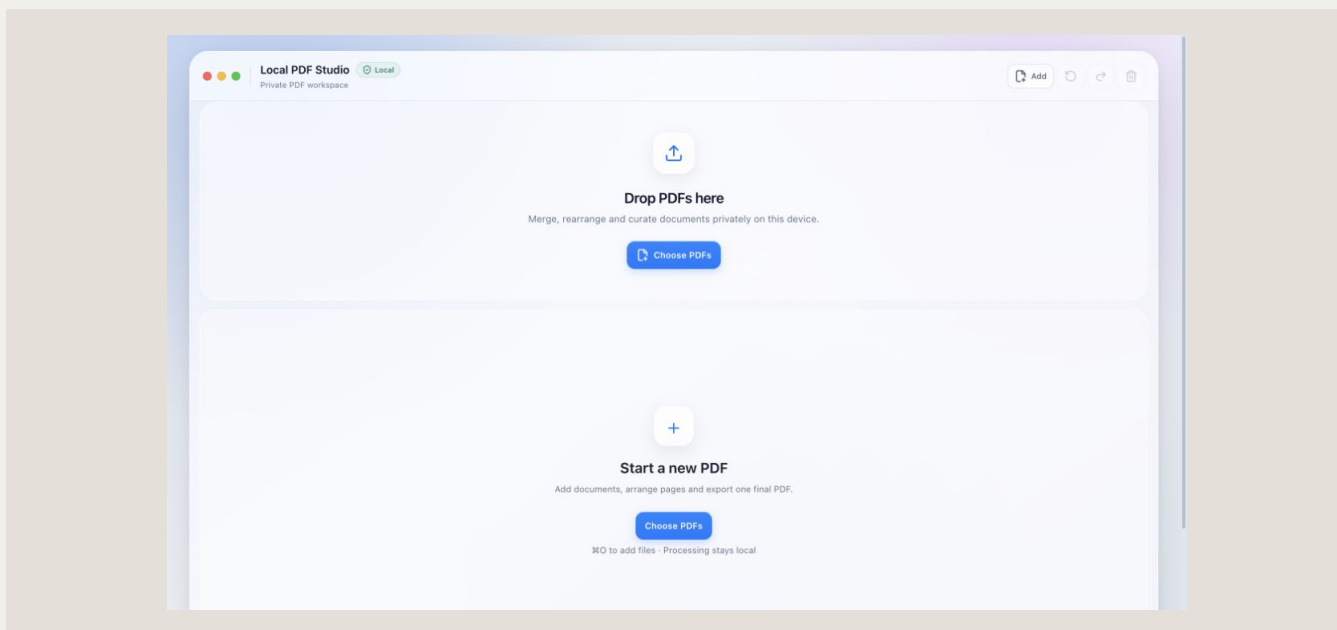
The screenshot is the original React dashboard rendered with synthetic emails. Source code exposes mailbox sync, keyword and semantic search, and evidence packaging. The preview does not demonstrate live Gmail OAuth or local-model accuracy. Some mailbox questions use dedicated data routes; keyword retrieval works without embeddings.

GlassPDF

Arrange, refine and export PDFs locally.

BROWSER APPLICATION

React / pdf-lib / PDF.js



Original interface rendered locally; empty workspace.

PURPOSE Give people direct control over PDF pages without uploading their documents to a processing server.

BUILD PDF.js renders previews; pdf-lib copies and exports pages. The workspace supports page and document ordering, rotation, duplication, ranges, and undo/redo.

EVIDENCE & SCOPE The repository separates rendering from export, caches parsed documents, and lazily renders thumbnails. The screenshot shows the original app, without private documents. Documents are held in browser memory. The repository does not provide saved workspaces, a processing backend or a multi-user collaboration service.

Inspect repository

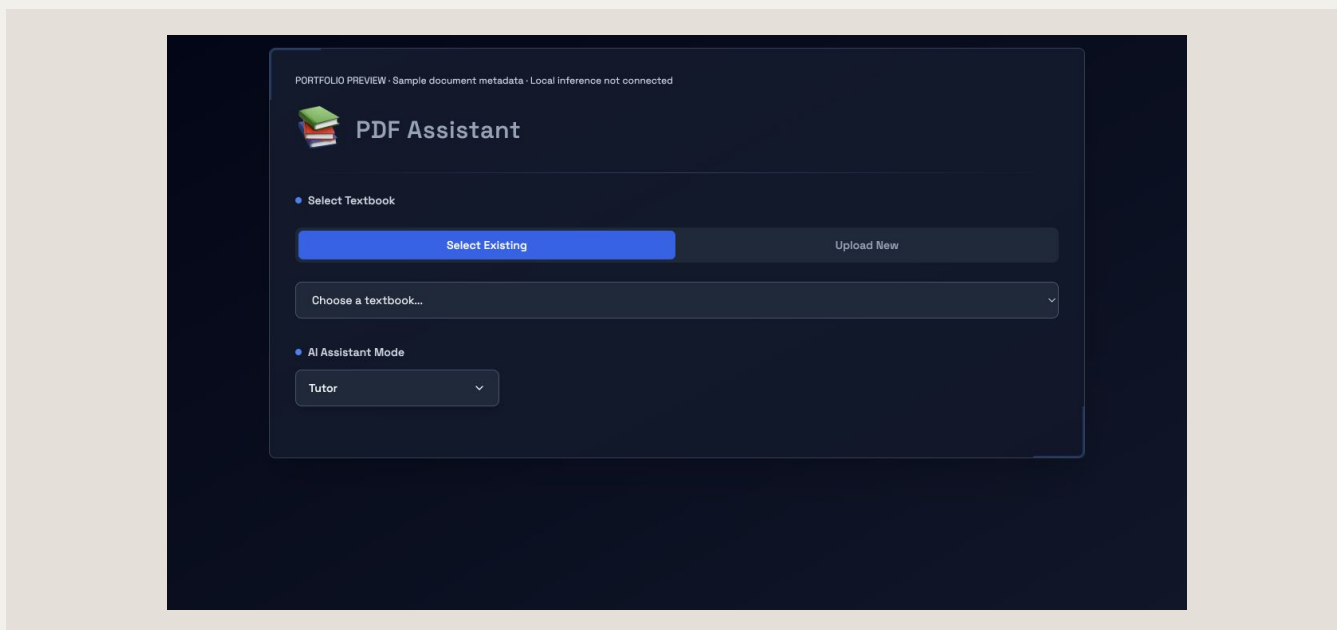
Reviewed source revision

DRAX TBS

Bring textbook context into a conversation.

LOCAL PROTOTYPE

FastAPI / ChromaDB / sentence embeddings / LM Studio



Original frontend with sample metadata; inference not connected.

PURPOSE	Retrieve relevant textbook passages before a local model responds.
BUILD	PyMuPDF extracts text into approximately 500-character chunks. Sentence embeddings are stored in ChromaDB; the query path retrieves eight chunks for LM Studio.
EVIDENCE & SCOPE	Source code covers ingestion, persisted vectors and context assembly. The original UI preview uses sample document metadata. The preview is not connected to inference. Retrieval context does not prove answer quality, and citation and response-mode limitations remain.

Teyrin.

Intelligence, put to work.

A software company
I'm building independently.

Solely owned by Nikhil Yarra.

Founder-led, from conversation to implementation.

01 Understand the need

Define the users, constraints and useful outcome.

02 Build the right scope

Develop the interface, data workflow and integrations.

03 Review and refine

Evaluate behavior, document boundaries and iterate.

[Teyrin on LinkedIn](#)

More to explore.

DaChat

CSV exploration and property prediction

Airspace

Distributed flight-data aggregation

Facial emotion recognition

Comparative CNN experiments

DDoS detection

Machine-learning security project

Let's build something useful.

A product, a workflow, or a role.
Start with a conversation.

nikhilyarra01@gmail.com

[Portfolio](#)

[LinkedIn](#)

[GitHub](#)

Project previews are demonstrations, not customer deployments.
